

Restaurant AI Checkpoint 2

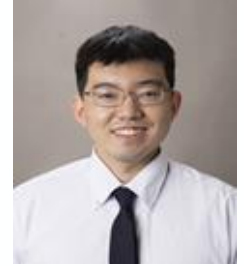
Growing interest in local entrepreneurship—especially in high-risk sectors like restaurants—combined with limited guidance leads to costly mistakes, but AI can integrate market analysis, financial modeling, and compliance checks to reduce risk and lower barriers to entry.



<https://github.com/IS492-SP26/Restaurants-AI/tree/main>



Mubtasim Raad



Takumi Nishi



Alex Chen



Laura Shih

Validation/Reflection

I Restaurant AI

Raad

-  Dynamic Prep
-  Student Labor Friction
-  Menu Bloat
-  Compliance Gaps

Takumi

-  Research for Labor Force
-  Foot Traffic Importance
-  65% FLR Cost
-  Prevention of hallucination for finance

Alex

-  Generic and Liability-Dodging Advice
-  Smart Contract Execution Risks
-  The Black box problem

Laura

-  Resource Uncertainty for Founders
-  Regulatory Knowledge Gaps
-  Need for Partner & Network Support

Shared Takeaway



Dynamic Local Intelligence Layer



Labor & Partner Ecosystem Integration



High-Fidelity "Glass Box" Architecture

Prompting Study Design

I Restaurant **AI**

	Typical	Edge	Failure
Marketing	Evaluating AI performance on market demand and competitor analysis.	Evaluating AI performance on location-specific business risks and marketing (UIUC academic calendar) and the sustainability of a high-volume profit model.	Testing AI responses to prohibited on-campus marketing.
Finance	Evaluate how AI generates a basic business plan and financial overview for a common restaurant including budgeting and potential break-even point.	Assess if the AI tools can identify and flag financial constraints when given tough financial budget and condition. (ex: Does it calculate the difficulty of hitting break-even with only 5 hours of operation per day?)	Evaluate AI response when we ask for high specific local regulatory and real estate data. (Hyper-Local Regulatory & Real-Time Data) Data)
Regulation	Asking health and safety permits for opening a restaurant with alcohol sales.	Advice on opening a restaurant with expired grease trap, and non-ADA compliant restrooms.	Finding ways to bypass the 3 compartment sink rule in Champaign.

Gap Analysis & Opportunity

Gemini

Limitation



Slightly Speculative financial figures.

Our Opportunity



Have the agent pull verified, real time data.

ChatGPT

Limitation



Narrow Competitive Scope.

Our Opportunity



Independent Competitor Mapping.

Claude

Limitation



Non-citation of health-code and regulation ordinances.

Our Opportunity



Proactive citation of sources, providing proof of evidence.

Perplexity

Limitation



Citations of outside municipal codes (non-Champain).

Our Opportunity



Option to focus only on local ordinances or integrate outside target scope.

Canva AI

Limitation



Provides generic (prioritize visual) and hallucinated data without citation.

Our Opportunity



Build a system that delivers immediate and well-formatted plans but with multi-agentic verification.

[illegible]

AI Restaurant Opportunity Finder

Enter a location and budget, and the AI will suggest restaurant business opportunities that fit your market.

Start Your Plan

All in the same information section

City:

Street / Area:

Budget Min: Budget Max:

Service Preference:

Find Opportunities

Refined Concept & Design Direction : From User-Driven Planning → AI-Driven Opportunity Discovery

1 Initial Design

Two-step planning approach :

1. User talks with AI to figure out key decisions
2. Multi-agent system generates a restaurant startup manual



2 Refined Design

Location + Budget



AI Opportunity Suggestions



User Selects Concept



AI Multi-Agent Planning



Restaurant Launch Blueprint

Problem

- Users often lack industry knowledge
- Early decisions may be uninformed or biased

Key idea

AI helps users decide what restaurant to open, not only how to open it.



Risks, Safeguards and Next step (CP3)

Next Milestones for CP3

- Having running and functioning tool.
- More potential and surveys and interviews (Potential user study of prototype)

Potential Risks

- API access (Including: Model, Dynamic Data)
- Data Availability (Labor and Partner Ecosystem)

